

A Generative Artificial Intelligence Based Tutor for Personalized Learning

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Abstract— Personalized learning refers to a system of teaching and learning where the content, methods, and assessment are tailored to each learner's needs, capacities/skills/competencies, and pace. There is abundant literature about personalized learning, describing its advantages, challenges, and approaches to implementing it. This study has explored the abundant literature and realized that while there are substantial proposed ways to achieve personalized learning, these proposals are not implemented into actionable products. This research aims to construct a Generative Artificial Intelligence (Gen AI) tutor that implements personalized learning and teaching in a higher education level course. The research methodology involved exploring literature to determine the requirements for personalized learning and designing a tutor system. The resultant system uses Custom GPT technology, Application Programming Interface (API), a repository for storing content, learners' profile information, and performance metrics. Out of the six components that make the proposed system, only the "Custom GPT" component has been implemented and tested on a course named "Applied Cryptography" in a postgraduate program. Though the preliminary results are promising, an effective assessment of the system will be made when the full implementation is completed. Further, issues related to ethics, scaling and integration with learning management systems should also be considered. Subsequent work to this study will focus on those issues.

Keywords— *Generative Artificial Intelligence, Personalized Learning, Artificial Intelligence Tutoring, Content Generation, Learners' profile.*

I. INTRODUCTION

Personalized learning has attracted a lot of interest from several educators and educational research. Though there is still no clear consensus definition of the term, it is generally accepted that personalized learning refers to tailoring learning to each learner's interest, strengths, and needs [1]. The approach of personalized learning encourages flexibility to support mastery and enables learners to actively take part in defining how, what, when, and where they learn [2]. According to [3], personalized learning is a student-centered approach that seeks to align the learning (content and method) to the students' individual needs and prior experiences. Even though the idea and interest of personalized learning is very old, and lot of efforts have been made to implement it, there is still no standard and accepted systems that implements effectively the concepts of the approach [4], [5]. This research aims to construct a Generative Artificial Intelligence (Gen AI) tutor that implements personalized learning and teaching in a higher education level course.

With the expansion of technology mediated learning, online and blended learning, the need for personalized learning is even more acute. A study conducted by Shemshack & Spector reveals that since 2008 there has been a significant increase in research and publications on personalized learning [6]. The recent advances in AI and specially Gen AI have increased the potential impact of technology on education leading to several studies investigating on how AI technologies can contribute to personalized learning [7], [8], [9]. More specifically, since the release of ChatGPT in November 2022, Gen AI has gained a lot of attention in education. Gen AI has been explored to address issues related to various aspects of personalized learning including adaptation to learners needs [10], learning plans [11], personalized dashboards through learning analytics [12], personalized assessment [13], and personalized interactions [14].

The assumption of this study is that by reducing the scope of the learning to a course level, it is possible to provide effective implementation of personalized learning using AI and Gen AI tools. After reviewing existing relevant literature, the study established the requirements of a system offering personalized learning. From the identified requirements, a design of such system was proposed. Three specific AI capabilities namely Natural Language Processing (NLP), Data Analytics (DA), and Computer Vision (CV) were leveraged to provide the desired features of the system. A partial implementation of the system has been done and tested, and the preliminary results are very promising.

The remaining part of the paper is organized into three sections. First the methodology section describes the research process, and the design of the proposed solution. Next, the results of the studies are presented and discussed. Finally, section V concludes the paper.

II. LITERATURE REVIEW

The review started by looking at literature related to personalized learning, then moves on to exploring the existing Gen AI tutors, and finally, a summary of the literature and existing gaps are established. The remaining part of this literature review section is organized around these three main points.

A. Personalized Learning

According to [15], personalized learning and teaching should be the direction to follow to provide the 21st century skills to learners making them fit for the ever changing job market. The author argues that the "one size fits all

approach” is no longer fit nor does it respond to the current needs and settings of learners. Exploring the important volume of existing literature, it is revealed that the implementation of personalized learning requires: availability of real-time data that build learner’s profile [2], structures for learners voice and feedback [15], [2], personalized pathways in the learning process [16], [17], the consideration of Universal Design for Learning framework (UDL) [18], scaffolding [19].

1) *Building learners’ profile*: Effective personalized learning cannot be achieved without having a proper mechanism for identifying the learner’s specifics here referred to as the learner’s profile. The key question is “what are the elements that form a learner’s profile in the context of personalized learning?” Analyzing existing literature, this study has established a list of 12 key elements summarized in Table I, along with the corresponding literature sources.

As [20] rightly puts it, personalized learning represents a teaching paradigm shift that reoriented the role of the teacher. The elements of Table I will be later further discussed and consolidated for implementation into a Gen AI tutoring system for personalized learning.

TABLE I. CHARACTERISTICS OF LEARNER'S PROFILE FOR PERSONALIZED LEARNING

#	Profile Elements	Corresponding Sources
1	Goal: Learning goal	[21], [22], [23]
2	Pace	[21], [22]
3	Time	[21], [22]
4	Task Strategy	[21]
5	Place	[22]
6	Performance Level	[3], [22]
7	Path	[22]
8	Learning Style	[24]
9	Strengths	[2], [23]
10	Needs	[2], [3], [23]
11	Motivations	[23]
12	Interest	[2]

2) *Learners Voice and Feedback*: Several studies including [3], [25] and [26] consider that true personalized learning must provide mechanisms for learners to voice their concerns and give valuable feedback useful to improve the teaching and learning. [3] argue that personalized learning environment provides mechanisms for learners’ voice and feedback enhances cognitive activation and increases support to learners.

3) *Personalized Pathways*: In learning, a pathway organizes knowledge in a way that shows a generalized version of the subject matter but with a flexible navigation [17]. Through the flexible navigation, personalized pathways allow each learner to adapt his or her own pathway to personal resources, timeframe, previous knowledge, needs and learning styles [17]. The learning style refers to how a learner perceives, gathers and processes

information; which is determined by a collection of individual skills and preferences [24]. Personalized pathways is therefore very central to personalized learning. Ingkavara et al. [21] argue that the ability to detect the learning style of a student enables to recommend the most appropriate learning material.

4) *Universal Design for Learning Framework*: Universal Design for Learning Framework (UDL) contribute to personalized learning by introducing three key principles which advocate that there are multiple ways of representation, expression and engagement [18]. The UDL framework also emphasizes assignment format (variety in assignment formats allow each learner to choose the format that works for him or her), self-monitoring and self-evaluation (the learner takes more responsibility in the learning process), and problem-based learning (encourages students to use their knowledge and skills to provide a solution to a given problem). In [2], the authors consider UDL as a scientific framework that supports variability of every learner through proactive and iterative design integrating multiple means of engagement, representation of information, and action and expression of understanding.

5) *Scaffolding*: In education, scaffolding refers to a timely and required assistance provided by a teacher or tutor to a learner, in solving an education problem, carrying out an education task, or achieve a goal that is beyond the knowledge, skills or competencies of the learner [27]. Scaffolding either by teachers or by peer-learners is essential in ensuring progress of learners towards their full potential. Scaffolding provides necessary encouragement for learning and should be considered as part of the design components of personalized learning [19]. According to Kubsch et al. [28], Scaffolding and feedback are very effective in providing individualized support and planning further instruction.

B. Artificial Intelligence Based Teaching Tutors

Lin et al. [7] argue that Artificial Intelligence (AI) can address some of the current challenges in education especially by creating personalized learning experiences. Intelligent tutoring systems with embedded AI can assist both students and teachers in providing personalized support [15]. In a study conducted by Benedict [29], it is concluded that Artificial Intelligence in Education (AIED) when combined with other forms of teaching is very effective specially for students in STEM areas of study. Depending on the underlying AI capabilities, the existing literature on AI teaching tutors with emphasis on personalized learning can be grouped into three categories: (1) tutors leveraging on Natural Language Processing (NLP), (2) tutors anchored on Data Analytics (DA) capabilities, and (3) tutors exploiting Computer Vision (CV) abilities [8].

Natural Language Processing (NLP). Using NLP, chatbot can be created for interaction, providing feedback, and making required learning interventions for learners [7]. NLP techniques are used in [9] to collect students comments or questions and generate high quality personalized feedback to students learning programming in Java. The results of this study revealed that NLP based approach has overcome the limitations of traditional rule-based approach

of generating feedback, which could not provide personalized feedback specific to the learner's needs. Another interesting application of NLP for personalized learning is demonstrated in [30] where the technique is used to group students by their shared interests. This is particularly useful for meaning interactions among peer students.

Data Analytics (DA). Machine Learning (ML) and other data mining techniques provide advanced DA capabilities which are useful for analysing learners' data (performance, progression, interaction, etc.) and make predictions that guide on how to adapt the learning to a learner [7]. The use of DA in learning has developed into a new field of study called Learning Analytics (LA) [31]. Using LA capabilities [32] proposed a method for tapping into the vast amount available learners' data acquire real-time insights of students' performance and adapt the learning path and content to avoid them falling behind. Building on DA, frameworks have been built to improve personalized learning environments by providing features such as visualizations, learning recommendations, reminders, self-assessments possibilities, and ratings [33]. In addition to supporting students, LA is also used to assist teachers in pre-service training as well [34].

Computer Vision (CV). Image recognition is a key capability of AI that enable AI-based tutors to identify facial expressions, gestures, and emotions to inform future actions in the context of personalized learning [7], [35]. CV is also used in active learning involving the use of a robot, where the robot must recognize and build a profile of the learner to customize its behaviour [36].

C. Summary and Gaps

In summary, personalized learning has been sufficiently explored, several studies have suggested list of issues to address, others have proposed how to address these issues. Among the proposed solutions, we have both non-AI based and AI-based solutions. However, through the extensive review of literature, this study identified the following two main gaps:

- Most proposed solutions do not provide concrete implementations that are available for use in real-world educational systems.
- The use of Gen AI to address personalized learning has not been adequately explored in the existing literature.

This study seeks therefore to leverage on Gen AI to provide a concrete useful solution for personalized learning in within the scope of a course. By reducing the scope to the course level, complexity is reduced, and the course level solution can later be scaled to a whole academic curriculum.

III. METHODOLOGY

This study is anchored on existing literature about personalized learning to determine the requirements of a tutoring system that supports personalized learning and teaching. The methodology uses a three-phase process sketched in Fig. 1.

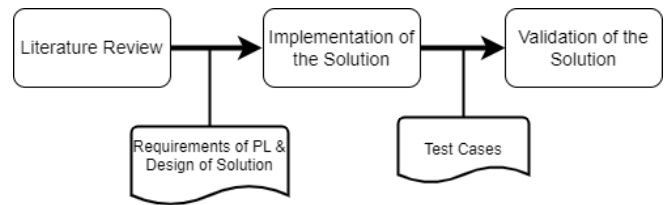


Fig. 1. Methodological Process of the study

- Phase 1 - Literature review: this phase is aimed at determining the requirements for personalized learning based on recommendations from previous studies. Next, based on the identified requirements, a design of a solution is proposed.
- Phase 2 – Implementation of the System: In this phase, the six components are incrementally implemented, starting with the Custom GPT.
- Phase 3 - Validation: In the validation phase, the components are individually tested, and next the system as a whole is evaluated.

The remaining part of this section describes in detail these three phases.

A. System Requirements and Design

Based on the review of literature, an efficient AI-based tutoring system requires NLP, LA and CV techniques and must offer the following features: (1) building and detecting learner profile, (2) storing and analyzing learner data (profile, performance, etc.), (3) detecting and analyzing learner's activities, emotions and behaviors, (4) recommending personalized content and activities to learners, and (5) providing an efficient interface for interactions of the learners with the system.

1) *Building and detecting Learner Profile:* Based on a consolidated view on the 12 keys elements of learner's characteristics identified in Table I, the learner profile information was established and related data are stored in a central data repository that holds all data handled within the system. The building of learners' profiles starts with a series of tests performed before the start of the course. The initial evaluation results are used to set the profile; but as the learner progresses into the course, the profile is updated accordingly. At any time, the system can access the learner's profile and provide feedback on necessary adjustments. Each time a learner logged into the system, the system read his/her profile data to detect the profile and provided necessary guidance, suggestions, and support.

2) *Storing and Analyzing Learner Data:* The data as indicated above, is stored in a central data repository which is accessible to the learning analytics engine responsible for the analysis. The analytics performed include learners' performance, learning time and pace, and progression.

3) *Detecting and Analyzing Learner's Activities, Emotions and Behaviors:* The system detects the learner's activities (reading course material, completing assignments, taking exam, requesting support, interacting with peer, interaction with teacher / instructor). In addition, the system makes use of the camera and other audio-visual devices available to detect the emotions and behaviors of the learner. Once the system has detected the activities and emotions of the learner, it can then in real-time analyse how that align

with the current learner's learning path. In doing so, deviations from promised learning paths are detected and the learner can be further guided.

4) *Recommending Contents and Activities*: Having access to the learner's profile data, the system can recommend content and activities to improve his/her performance. These recommendations can be provided automatically in real time, or they can be given upon the learner's request. The recommendations can cover areas spanning from content recommendations, path recommendations, or activities recommendations.

5) *Providing Interface for Interactions*: The success of this complex tutoring system depends on the provision of a comprehensive and intuitive user interface. The traditional prompting interface of Chat tools will be replaced with an elaborate user interface where the learner has more options to ask questions and get responses. The system then internally transforms the complex user input into a well-crafted prompt message to be given to the GPT models.

Considering the five requirements above, a high-level description of the proposed solution with seven main components is provided in Fig. 2.

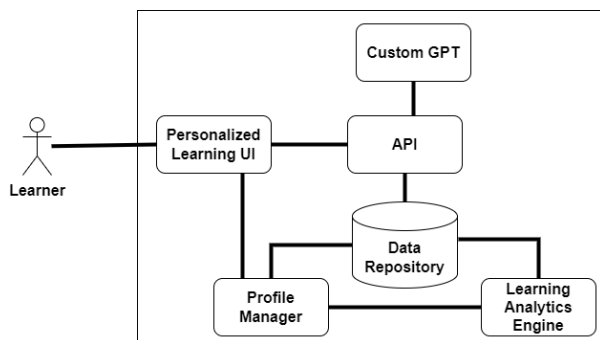


Fig. 2. Design of the proposed solution

The six main components are described as follows:

- **Custom GPT**: This is a ChatGPT based component which has been retrained with the course material. All resources related to the course are used for the retraining so that the resulting customize chat can respond with specific knowledge related to the course. It is also in charge of content generation
- **API**: It allows all the other components of the system to access the Custom GPT using API integration.
- **Personalized UI**: The traditional text box and prompting approach of interaction with chatbots in this case ChatGPT is not adequate for the purpose of personalized learning for two reasons: (1) it requires learner to master prompt engineering to be able to craft good prompts to get the most appropriate answers. (2) in personalized learning, audio-video devices available must be used to detect the learner's emotions and provide support accordingly. For these reasons, the proposed solution integrates a specialized UI.
- **Profile Manager**: The learner profile is very central in personalized learning. This module is responsible for detecting and updating the learners' profiles. As indicated in Fig. 1, it is connected to

the data repository (to access to the learners' data), to the learning analytics engine (to obtain relevant analytics of learners), to the personalized UI (to provide relevant content and UI elements).

- **Data Repository**: This module manages all the data of the system, meaning the learners' profiles data, some generated content, and data related to the configuration of the system.
- **LA Engine**: As indicated in the list of requirements for personalized learning, this module analyses the learners' performances, their pace, time, and progression. These analytics can assist the profile manager to suggest relevant learning paths or specific content or assignment to the learners.

B. Implementation of the System

The implementation aims at writing the code for all the six modules and their interconnections. From the user perspective, the implementation will provide a web interface through which learners will access the system. The approach is to reuse as much as possible existing software components to minimize the amount of code. The Python programming language is selected for the implementation because of its extensive libraries and its versatile use in AI coding. Specific implementation details for the six main components are given as follows:

- **Custom GPT**: This component is implemented using the ChatGPT customization methods. The model is further trained on the specific course material contents.
- **Personalized UI**: This module has two main elements: (1) the UI which is a web interface implemented using python web development tools. (2) a sensing unit for capturing video and sound from the learner, enabling to detect the possible emotions of the learner. Python-opencv [37][38], python-sounddevice [39] and pyaudio [40] libraries are used to implement this unit.
- **API**: The unit is implemented in Python, realising various API calls and returning results to the calling components.
- **Profile Manager**: The implementation of the module uses a mix a rule-based system and machine learning algorithms.
- **Data Repository**: The repository is made of two units: (1) the operational data which is implemented as a MongoDB database [41], enabling the management of large volume of data comprising of both structured and non-structured data. (2) a mini data warehouse for the analytics purposes. Standard data warehousing technologies are used to implement this unit.
- **Learning Analytics Engine**: Machine Learning techniques, and other data mining techniques like association rules are used to implement this module.

In summary, the implementation is a Python based using web technologies, custom GPT technologies, computer vision and machine learning techniques, MongoDB database

and data warehouse technologies. Only the “Custom GPT” module has been implemented. The implementation of the other modules is ongoing.

C. Validation

To validate the proposed solution, the testing approach has been selected. To test the system, a course is selected, and a group of 12 research participants are selected to represent the learners. Using a purposive sampling, the 12 participants are selected in a way to have several learner profiles in the group. Two types of testings are used in this study: requirements testing, and user satisfaction testing.

1) *Requirements Testing:* The aim of this testing is to assess how much the five requirements described in section II.B are fulfilled by the system. For each of these requirements, a set of test cases is elaborated and tested with the system and the group of 12 research participants acting as the learners. The learners access to the system and start using the system, performing different types of learning activities, then direct observation is made on the system behavior and compared against expectations. A scoring system expressed percentage is used to determine the performance level of the system for each of the requirements. Performance level of 80% will be used to validate a requirement.

2) *User Satisfaction Testing:* This type of testing is about determining how the system is useful, meaning that it is actually helping the learner. For this testing, the course will be divided into 8 modules, and the 12 learners are divided into two groups numbered 1 and 2. For each module of the course, one group has access to the system and the other group is not allowed to use the system. Then we

observe the performance of the learners in the module. In particular we determine if there is a significant difference between the results of learning using the system and those of the group not using the system. In the next module of the course, we alternate the groups in terms of access or not to the system. By the completion of the eight modules of the course, we are able not only to measure the impact of the use of the system on learners’ performance, but we can also establish if there is consistency in the impact of the use of the system from module to module.

IV. RESULTS AND DISCUSSIONS

A. Results

The objective of this study was to create a Gen AI based tutor for personalized learning. Through an extensive literature exploration, the requirements of such a system were established, and a design was proposed. The implementation of the system has started. The Custom GPT module out of the six modules was entirely implemented, while the remaining ones are still in development. The implemented Custom GPT has been tested as a separate component. The testing was done with a graduate course entitled “Applied Cryptography”, and the aim of the testing was to determine whether the “Custom GPT” provides better answers compared to ChatGPT. Table II gives a sample of ten questions and the responses of the “Custom GPT” and “ChatGPT” showing that in 3/5 times of the cases, the Custom GPT provides better responses than ChatGPT. Another significant result of this study is the elaboration of a clear process to validate the outcome of the study.

TABALE II. COMPARISON BETWEEN CUSTOM GPT AND CHATGPT

#	Question	Expected Answer	Custom GPT	ChatGPT	Custom GPT Better?
1	What are the components of a cryptosystem?	The three main goals of computer security referred to as the CIA Triad are: Confidentiality, Integrity, and Availability.	The three main goals of computer security are commonly represented by the CIA Triad, which stands for: 1. Confidentiality: Definition: Ensuring that sensitive information is accessible only to those authorized to view it. Confidentiality prevents unauthorized disclosure of information. Examples: Encryption is a technique used to maintain confidentiality by transforming readable data into an unreadable format, accessible only by those possessing the correct decryption key. Access control mechanisms such as passwords and user authentication are also used to maintain confidentiality. 2. Integrity: Definition: Ensuring that information remains accurate, complete, and unaltered except by authorized individuals. Integrity involves guarding against improper modification or destruction of information. Examples: Hash	The three main goals of computer security are commonly referred to as the CIA triad, which stands for: 1. Confidentiality: Ensuring that sensitive information is accessible only to authorized individuals and is protected from unauthorized access or disclosure. This involves implementing measures like encryption, access controls, and authentication protocols to safeguard data privacy. 2. Integrity: Maintaining the accuracy and completeness of data throughout its lifecycle. Integrity ensures that information is not altered or tampered with by unauthorized parties. Techniques such as checksums, hashing, and digital signatures are used to detect and prevent unauthorized modifications. 3. Availability: Guaranteeing that information and resources are accessible to authorized users whenever needed. This	NO

			functions and message authentication codes (MACs) are used to verify that data has not been tampered with during storage or transmission. Digital signatures also provide a way to ensure data integrity by enabling verification of the source and content of a message. 3. Availability: Definition: Ensuring that information and systems are available and accessible to authorized users when needed. Availability aims to prevent disruptions to services and data accessibility. Examples: Techniques like redundancy, failover, backups, and Denial-of-Service (DoS) attack mitigation are employed to ensure the availability of systems and data. Maintaining server uptime, load balancing, and implementing measures against network attacks are all part of achieving availability.	involves protecting systems against disruptions such as hardware failures, cyber-attacks (like Denial-of-Service attacks), and other events that could render data or services unavailable. Redundancy, failover systems, and regular maintenance are strategies used to enhance availability.	
2	Is RSA a symmetric or asymmetric system?	RSA is an asymmetric cryptographic system.	RSA is an asymmetric cryptographic system, also known as a public-key cryptosystem	RSA is an asymmetric cryptographic system.	NO
3	What is authentication?	Authentication is the process of ensuring that the receiver is able to ascertain the origin of the message. An intruder should not be able to masquerade as someone else.	The receiver must be able to ascertain the origin of the message. An intruder should not be able to masquerade as someone else.	Authentication is the process of verifying the identity of a user, system, or entity before granting access to resources or services. It ensures that someone or something is who or what it claims to be. This verification is a fundamental aspect of security in both physical and digital environments, serving as the first line of defense against unauthorized access.	YES
4	What is information theory?	Information Theory is the science that studies the coding, quantification, storage, and communication of digital information.	Information Theory is the field of study concerned with the coding, quantification, storage, and communication of digital information. It was first formalized by Claude Shannon in 1948, laying the groundwork for understanding how information can be transmitted and processed efficiently. It serves as a theoretical basis for many areas of cryptography.	Information Theory is a branch of applied mathematics and electrical engineering that deals with quantifying, storing, and communicating information efficiently and reliably. Introduced by Claude E. Shannon in his groundbreaking 1948 paper "A Mathematical Theory of Communication," it provides a mathematical framework for understanding the transmission, processing, and utilization of data.	YES
5	What are the three types of cryptographic protocols	The three types of cryptographic protocols are arbitrated, adjudicated, and self-enforcing protocols	The three types of cryptographic protocols are: 1. Arbitrated Protocols: These protocols involve a trusted third party, known as an arbitrator, who oversees and helps complete the transaction between the participating parties. The arbitrator's role is crucial to ensure that both parties fulfill their obligations without needing to trust each other. 2. Adjudicated Protocols: In these protocols, a third party may intervene in case of a dispute. Unlike arbitrated protocols, the third party (referred to as an adjudicator) is	The three main types of cryptographic protocols are: 1. Encryption Protocols: These protocols ensure the confidentiality of data by transforming readable information (plaintext) into an unreadable format (ciphertext) using encryption algorithms. Only authorized parties with the correct decryption key can revert the ciphertext back to plaintext. Examples include Secure Sockets Layer (SSL)/Transport Layer Security (TLS) and Internet Protocol Security (IPsec). 2. Authentication Protocols: These protocols verify the	YES

			not directly involved unless something goes wrong or there is a need for mediation or resolution. 3. Self-Enforcing Protocols: These protocols are designed so that both parties can verify compliance on their own without needing a third party. They are "self-enforcing" because the protocol itself ensures that the parties fulfill their obligations, making it unnecessary for an external arbitrator or adjudicator to be involved.	identity of users or devices before granting access to resources or information. They ensure that entities are who they claim to be, preventing unauthorized access. Common authentication protocols include Kerberos, Challenge-Handshake Authentication Protocol (CHAP), and Extensible Authentication Protocol (EAP). 3. Key Exchange Protocols: These protocols enable secure sharing of cryptographic keys between parties over an insecure channel. The keys are then used for encryption and decryption in secure communications. Key exchange protocols ensure that even if an attacker intercepts the communication, they cannot derive the keys being exchanged. Examples include the Diffie-Hellman key exchange and Elliptic Curve Diffie-Hellman (ECDH).	
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B. Discussions

The merit of this study compared to previous related studies resides in two elements: (1) going beyond suggestions of theories and concepts to propose a concrete tutoring system that effectively meets the requirements of personalized learning. (2) the usage of Gen AI to address personalized learning implementation. While the approach used in this research is promising to yield a good system for personalized learning, three key issues must be addressed to make the system even much better. These issues are ethical considerations, scaling, and integration with existing learning management systems (LMS).

1) *Ethical Considerations*: The use of Gen AI in education has often raised ethical concerns. In general institutions of learning have not developed policies and legal frameworks around the use of Gen AI [42]. It is advisable to any institution wishing to use the proposed system to develop a minimum Gen AI use policy so that it can be taken into consideration in the deployment and customization of the system. Another concern in the use of Gen AI is the reliability; how much can human rely on the output of AI tools [43]? In the proposed solution, the GPT models are retrained on specific content and therefore the tool is likely to be less free of hallucinations, but it is not guarantee to be 100% accurate.

2) *Scaling*: The proposed solution has some cost implication that can be a barrier to a large adoption of the system. 1) The customization of ChatGPT requires a subscription. 2) the use of APIs also has financial cost. While the subscription cost can be negligible, cost for using APIs is based on the volume of requests, and can lead to significantly high amounts if used by a large number of learners for many courses within an institution of learning. Nevertheless, it is reasonable to assume that in the near future, free and good APIs might be made available for GPT models.

3) *Integration with Learning Management System*: Most learning institutions have LMS; for convenience, it is recommended that any other learning system must integrate with the LMS. The proposed tutoring system must take into consideration the integration with standard and well known LMSs for more convinency for the learners.

V. CONCLUSION

Through a rigorous literature review, this study has identified the key elements and requirements for personalized learning implementation. Based on these elements and requirements, a design of a Gen AI tutor for personalized learning was proposed. Currently the system is partially implemented and results of initial testing of the implemented part are promising.

The main contribution of this study is summarised in the following three points: 1) the elaboration of the requirements for implementing personalized learning, 2) the design of Gen AI tutor for personalized learning, and 3) the establishment of a clear methodology for validating the system by testing.

Though a significant step has been made in this paper towards implementing a Gen AI based tutor for personalized learning, there is still a long way to completing the implementation of the remaining modules of the system. Furthermore, as mentioned in the discussions section, ethical considerations, scaling and integration of existing LMS are important issues that must be factored in the future works.

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